Paper Notes — DemoGraft (working title)

Planning doc for CoRL submission: **1 demo → many cousin demos → robot policy training**.

Story

> Given **one RGB video** of a human performing a task (e.g. picking up a mug, placing it on a coffee machine, opening a drawer) with the reference objects already scanned into Gaussian splats, automatically generate N × M cousin demos across visually-distinct variant objects via semantic-aligned 3DGS transfer. Train a robot policy on the synthesized set — scalable "free" data augmentation for robotic manipulation.

Single thesis: **zero-shot demo augmentation via semantic-distilled 3DGS alignment, seeded by a single RGB video**. Not N sub-pipelines — one meta-pipeline with a crisp input (1 video) and output (many demos).

End-to-end flow

```
1 RGB video                 → track_with_dino_sam3.py  → per-frame SE(3)
+ reference splats          (DINO + RGB + silhouette + depth losses)
+ scanned cousin splats     → Segmentation (SAM3) + Alignment (semantic
                            → Motion transfer (prismatic R^T·a, revolute
                            → IsaacSim / real-robot replay
                            → Policy training (DP / ACT / π0)
```

Why it works

- **RGB-only input:** no depth camera needed at demo time — phone video, web video, old archive footage all work. Massively lowers the barrier vs BundleSDF/FoundationPose-style pipelines.
- **Semantic correspondence, not geometric:** DINO-weighted ICP matches drawer-front to drawer-front even across wildly different cabinet bodies. Extracted parts and transferred axes stay anchored to the same functional regions.
- **Zero per-cousin annotation:** text prompt + one-time reference motion covers the whole cousin set.
- **Preserves contact constraints:** region-weighted semantic ICP biases alignment toward mug-contact areas on the coffee machine.
- **Unified framework:** rigid placement (mug-on-CM) and articulation (drawer open) fall out of the same pipeline. Not kitchen-sink — unified.

Differentiation from prior art

Related	How we differ
Video-retarget (Gen2Act, VideoMimic)	They retarget human motion; we retarget object placement + articulation

Video → pose w/ depth (BundleSDF, FoundationPose, StablePose)	**They require a depth sensor (RGB-D).** We use **RGB only —** phone video or web video works. Depth-ranking loss from DepthAnything V2 substitutes for a depth camera.
Video → pose RGB-only (RSRD)	They render the full scene incl. background; we render only the object splat and rely on sharpened-silhouette + distilled-DINO losses. Our splat is pre-distilled with DINO — RSRD uses CLIP; our 8-seed multi-start init handles larger pose ambiguities.
Demo augmentation (MimicGen, RoboCasa, DemoGen)	They require per-object engineering; we zero-shot swap via DINO-aligned splats
DINO-lifted splats (F3RM, LangSplat, GARField, DIG)	They do retrieval/segmentation; we use the features for cross-object motion transfer
Screw-axis from video (ScrewSplat, Ditto, PARIS, CARTO)	They estimate axis on a single object; we transfer axis across cousins via `R^T`
Real2Sim2Real (RoboCasa)	General framework; our novelty is the semantic-only cross-object motion transfer

Technical contributions (what's novel)

1. **RGB-only video → splat pose tracking with distilled DINO features** (`track_with_dino_sam3.py`). Per-frame SE(3) optimization of a reference Gaussian splat against video frames via distilled-DINO + RGB + silhouette + DepthAnything-V2 depth-ranking losses. **No depth sensor required** — works on any phone video (unlike BundleSDF/FoundationPose which require RGB-D). 8-seed multi-start init handles pose ambiguity. Produces the trajectory that seeds the whole augmentation pipeline — no teleop needed.
2. **Cross-object motion transfer via semantic 3DGS alignment** — clean `R^T · a_src` math for prismatic, `(R·ω, R·(p·s)+t)` for revolute. Translation drops out for direction-only motion. Scale composes multiplicatively.
3. **Region-weighted contact-aware alignment** — click-to-weight semantic ICP with DINO-propagated source+target masks biases alignment toward contact regions. Not standard in real2sim.
4. **Part-to-part alignment with `--no-center`** — aligns extracted subparts while preserving parent-frame coords, so the same transform aligns the cousin's whole scene.
5. **Unified rigid-placement + articulation pipeline** — one framework handles both.

Candidate names

Name	Why	Grade
DemoGraft ☆	biological metaphor (grafting a shoot onto a new tree), memorable, captures "one → many"	1st
SplatCast	medium-specific, punchy, action verb	2nd
CousinCast	literal, ties to our "cousin" framing	3rd
DemoGen3D	ties to demo-gen lineage, claims 3D novelty	alternative

Working title: **DemoGraft: One Demo, Many Objects — Semantic 3D Gaussian Splat Alignment for Scalable Robot Demo Synthesis**

Experiments to nail for CoRL

Required

1. **Hero task:** single real-robot task combining both rigid placement and articulation. Suggested: *open drawer → place mug inside → close drawer* OR *place mug on coffee machine → press button*. Pick one, nail it.
2. **Cousin counts:** aim 1 → 50+ cousin demos. Report exact numbers (K mugs × L CMs × J drawers).
3. **Policy headline metric:** success rate of a policy (diffusion policy / ACT / similar) trained on the synth set.
4. **Generalization test:** held-out mug or coffee machine at real-robot test time.

Baselines

- **(a)** Policy trained on 1 original demo (lower bound)
- **(b)** Policy trained on N real teleop demos, same human-time budget (apples-to-apples)
- **(c)** MimicGen-style demo augmentation on the same cousin set
- **(d)** RoboCasa / DemoGen-equivalent if applicable

Ablations

- No semantic alignment (pure ICP) — shows DINO features matter
- No region-weighted refinement — shows contact-aware alignment matters
- No ground-plane snap — shows base-plane alignment matters
- Whole-scene vs part-to-part alignment (`--no-center` trick) — shows segmentation-first matters
- No scale composition (fix `s=1`) — shows scale matters for size-variant cousins

Scaling plot

- X-axis: # cousin demos (1, 5, 10, 20, 50)
- Y-axis: real-robot success rate
- Show saturation curve, confidence intervals

CoRL tier assessment

Factor	Status	Notes
Technical novelty	Strong	Semantic-only motion transfer + unified framework are genuine
Infrastructure built	✓	Segmentation, alignment, motion transfer, viewers all exist
Clear single thesis	✓	"1 demo → many cousins"
Scaling story with numbers	**Gap**	Need to run 50+ cousin demos
Head-to-head baselines	**Gap**	Need to run MimicGen/RoboCasa on cousin set
Real-robot success rates	**Critical gap for oral**	Must have

Held-out generalization	**Gap**	Required for strong paper

Realistic tier outlook

- **Poster:** highly likely given current story + novelty.
- **Highlight/Spotlight:** very plausible with real-robot numbers + scaling plot + 2 baselines.
- **Oral:** requires numerical wins over MimicGen/RoboCasa on the same tasks *and* unified rigid+articulation with real robot. Doable but tight — ~6-8 weeks of solid experiment execution.

What pushes poster → oral

1. **Scaling plot** showing saturation — 1 → 50+ cousin demos.
2. **Head-to-head numbers** beating MimicGen or RoboCasa on the same cousin set.
3. **Unified real-robot task** requiring both rigid and articulation (drawer + mug).
4. "**Works on visually wild cousins**" story — DINO doing heavy lifting across large appearance gaps.

Built infrastructure (already done)

Pipelines

- **Video → pose tracking** (`scripts/track_with_dino_sam3.py`, see `scripts/TRACKING_METHOD.md`). SAM3 text mask + DINOv3 + RGB + silhouette + depth-ranking losses; 8-seed multi-start init; 60 iters/frame; outlier rejection + optional temporal smoothing. Output: `tracked_poses.npy` (N, 7) + debug video.
- DINOv3 distillation into 3DGS (`train_dinov3_features.py`)
- SAM3 text-prompted part segmentation (`segment_<object>.sh` — drawer, laptop, faucet, toaster)
- Multi-view mask voting extraction (`extract_part_gaussians.py`)
- Complement extraction (`extract_complement.py`)
- Part-to-part alignment with `--no-center` (`align_drawer_2_to_1.sh` pattern)
- Region-weighted refinement (`refine_<pair>.sh` pattern)
- Ground-plane alignment (`view_aligned_mugs_ground.py`)
- Prismatic motion authoring + transfer (`view_drawer_articulated.py`, `view_cousin_articulated.py`)
- Revolute (screw-axis) motion estimation + transfer (ScrewSplat, `transfer_and_articulate.py`)
- Cross-object + scene-level placement (`place_mug_on_cm.sh`)

Data assets (2026-04-21)

- Mugs: mug_1, mug_2, mug_3, sam_mug1, sam_mug2 (5)
- Coffee machines: cm_1, cm_2, cm_4, cm_8 (4)
- Drawers: drawer_1, drawer_2 (2 — need more)
- Faucets: faucet_1, faucet_2, faucet_n (3)
- Laptops: asus, kawhi, laptop1, laptop2, design_laptop (5+)

Saved transforms

- `mug_X_to_mug_2_aligned.pt` across all variant mugs
- `mug_X_in_cm_Y.pt` for cross-scene placement
- `cm_N_to_cm_1_aligned.pt` for cross-CM alignment with mug-region weighting
- `drawer_2_to_drawer_1_aligned.pt` for cousin drawer articulation

Open questions / to-do

- [ ] Select hero real-robot task
- [ ] Scale up cousin mugs (currently 5 → target 20+)
- [ ] Scale up cousin coffee machines (4 → 10+)
- [ ] Scale up cousin drawers (2 → 10+)
- [ ] Pick policy architecture (diffusion policy vs ACT vs π_0)
- [ ] Implement MimicGen baseline on the same cousin set
- [ ] IsaacSim task setup for the unified pick-place-with-articulation task
- [ ] Real-robot cell setup (UR5? Franka?)
- [ ] Identify 2-3 held-out cousin objects for generalization tests
- [ ] Stress-test visually wild cousins (unusual colors, shapes, logos)
- [ ] Ablation suite scripting